Introduction To Flight Mechanics

Tutorial 12



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Exercise: Basic Controller for the Lateral Motion

The following state space model of a generic aircraft (Class II) in horizontal cruise flight with $V_0 = 150\frac{m}{s}, h = 1000m$) is given:

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} -0.87 & 6.47 & -0.41 & 0 \\ -1 & -0.38 & 0 & 0.07 \\ 0.91 & -18.8 & -0.65 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} -0.88 & -7.78 \\ 0 & 0.12 \\ -14.3 & 3.74 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$

a) Determine the control structure and the corresponding controller gains such that the Roll Mode of the aircraft fulfills level 1 handling qualities according to MIL-F-8785C.
b) Determine the handling qualities level according to MIL-F-8785C for the Dutch Roll.
c) Propose a controller structure that changes the relative damping of the Dutch Roll.
d) Determine the gain which can be used to achieve a general, desired relative damping. You can assume that the eigenfrequency of the Dutch Roll remains unchanged by the feedback control.

Determine the structure and the gains of a controller, such that the **roll mode** of the aircraft satisfies **level 1 handling qualities** according to **MIL-F-8785C**.

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} -0.87 & 6.47 & -0.41 & 0 \\ -1 & -0.38 & 0 & 0.07 \\ 0.91 & -18.8 & -0.65 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} -0.88 & -7.78 \\ 0 & 0.12 \\ -14.3 & 3.74 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$



Note down the roll mode state space model.

Determine the structure and the gains of a controller, such that the **roll mode** of the aircraft satisfies **level 1 handling qualities** according to **MIL-F-8785C**.

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} -0.87 & 6.47 & -0.41 & 0 \\ -1 & -0.38 & 0 & 0.07 \\ 0.91 & -18.8 & -0.65 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} -0.88 & -7.78 \\ 0 & 0.12 \\ -14.3 & 3.74 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$



Note down the roll mode state space model.

$$\begin{bmatrix} \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} L_p & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} p \\ \Phi \end{bmatrix} + \begin{bmatrix} L_\xi \\ 0 \end{bmatrix} \cdot \xi$$

$$L_p = -0.65$$

$$L_\xi = -14.3$$

Table from MIL –F 8785

MIL-F-8785C

3.3.1.2 Roll mode. The roll-mode time constant, τ_R, shall be no greater than the appropriate value in table VII.

Maximum Time Constant Roll Mode in s

Phase	Class	Level		
		1	2	3
A	I, IV	1,0	1,4	10
	II, III	1,4	3,0	
B	All	1,4	3,0	10
C	I, II-C, IV	1,0	1,4	10
	II-L, III	1,4	3,0	

According to exercise:

- generic aircraft (Class II)
- in horizontal cruise flight
- level 1 desired

The MIL-F-8785C defines four aircraft types:

Class I

Small, light airplanes

- Light utility
- Primary trainer
- Light observation

Class II

Medium-weight, low-to-medium maneuverability airplanes

- Heavy utility/search and rescue
- Medium Transporter
- Early warning/ electronic counter-measures/airborne command, control, or communications relay
- Antisubmarine
- Assault Transport
- Reconnaissance
- Tactical bomber
- Heavy Attack
- Trainer for Class II

Class III

Large, heavy, low-to-medium maneuverability airplanes

- Heavy transporter/ cargo/tanker
- Heavy bomber
- Trainer for Class III

Class IV

High-maneuverability airplanes

- Fighter/Interceptor
- Attack
- Tactical reconnaissance
- Observation
- Trainer for Class IV

Nonterminal Flight Phases

Category A

Require rapid maneuvering, precision tracking, or precise flight path control

- Air-to-air combat (CO)
- Ground attack (GA)
- Weapon delivery/launch
- Aerial recovery (AR)
- Reconnaissance (RC)
- In-flight refueling (receiver) (RR)
- Terrain following (TF)
- Antisubmarine search (AS)
- Close formation flying (FF)

Category B

Normally accomplished using gradual maneuvers and without precision tracking, although accurate flight-path control may be required

- Climb (CL)
- **Cruise (CR)**
- Loiter (LO)
- In-flight refuelling (tanker) (RT)
- Descent (D)
- Emergency descent (ED)
- Emergency deceleration (DE)
- Aerial delivery (AD)

Terminal Flight Phases

Category C

Normally accomplished using gradual maneuvers and usually require accurate flight-path control.

- Takeoff (TO)
- Catapult takeoff (CT)
- Approach (PA)
- Wave-off/ go-around (WD)
- Landing (L)

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According to exercise:

- generic aircraft (Class II)
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According to exercise:

- generic aircraft (Class II)
- in horizontal cruise flight
- level 1 desired

Do we need a controller at all?

Check T_R

Determine T_R
Roll mode shows PT1 behavior:

$$\begin{bmatrix} \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} L_p & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} p \\ \Phi \end{bmatrix} + \begin{bmatrix} L_\xi \\ 0 \end{bmatrix} \cdot \xi$$

Equation from 2x2-System

$$\dot{p} = L_p \cdot p + L_\xi \cdot \xi$$



Derive the transfer function from ξ to p. What is the time constant of the resulting linear first order system (PT1)?

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Roll mode shows PT1 behavior:

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Derive the transfer function from ξ to p. What is the time constant of the resulting linear first order system (PT1)?

Laplace tranform: $s \cdot p = L_p \cdot p + L_\xi \cdot \xi$

$$(s - L_p) \cdot p = L_\xi \cdot \xi \quad \Longrightarrow \quad \frac{p}{\xi} = \frac{L_\xi}{(s - L_p)} = \frac{L_\xi}{-L_p} \frac{1}{\left(-\frac{1}{L_p}s + 1\right)} = \frac{L_\xi}{-L_p} \frac{1}{(T_R s + 1)}$$

$$T_R = -\frac{1}{L_P} = \frac{1}{0.65} = 1.54$$

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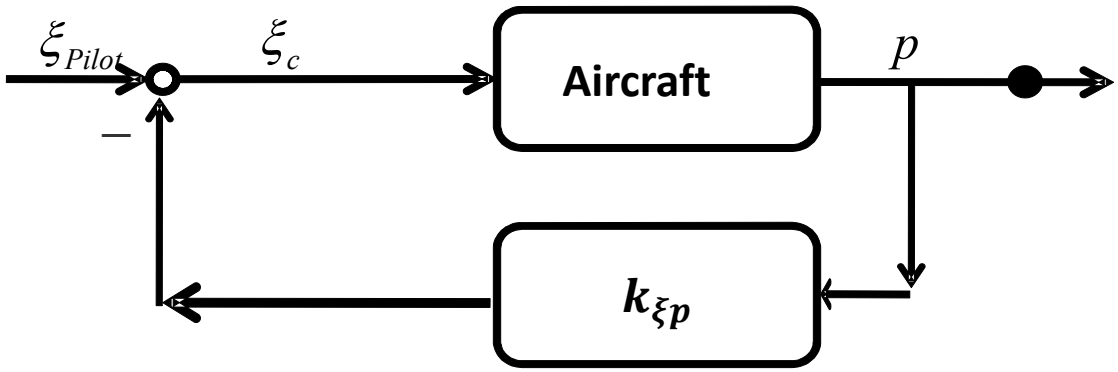
According to exercise:

- generic aircraft (Class II)
- in horizontal cruise flight
- level 1 desired

Do we need a controller at all?

Check T_R ⟹ $T_R = -\frac{1}{L_P} = \frac{1}{0.65} = 1.54 > 1{,}4$ ⟹ Controller needed

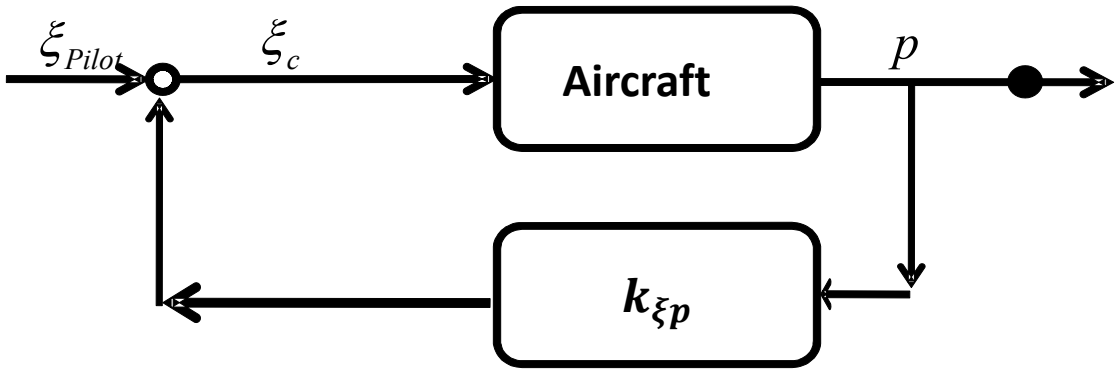
Controller structure: Feedback of the roll rate p



Control Law: $\xi_c = \xi_{Pilot} - k_{\xi p} \cdot p$

Derive the closed loop dynamics: $\dot{p} = L_p \cdot p + L_\xi \cdot \xi_c$

Controller structure: Feedback of the roll rate p



Control Law: $\xi_c = \xi_{Pilot} - k_{\xi p} \cdot p$

Derive the closed loop dynamics: $\dot{p} = L_p \cdot p + L_\xi \cdot \xi_c$

$$\dot{p} = L_p \cdot p + L_\xi \cdot \left(-k_{\xi p} \cdot p + \xi_{Pilot}\right)$$

$$\dot{p} = \left(L_p - k_{\xi p} \cdot L_\xi\right) \cdot p + L_\xi \cdot \xi_{Pilot}$$

$$\dot{p} = L_p^* \cdot p + L_\xi \cdot \xi_{Pilot} \qquad L_p^* = L_p - k_{\xi p} \cdot L_\xi$$

Determine $k_{\xi p}$ via artificial desired L_P^*:

$$L_P^* = L_p - k_{\xi p} \cdot L_\xi$$

$$T_R^* = -\frac{1}{L_p^*} = -\frac{1}{L_p - k_{\xi p} \cdot L_\xi} \overset{!}{=} T_{R,des}^*$$

$$-1 = T_{R,des}^* \left(L_p - k_{\xi p} \cdot L_\xi\right)$$

$$T_{R,des}^* k_{\xi p} \cdot L_\xi = T_{R,des}^* L_p + 1$$

$$k_{\xi p} = \frac{T_{R,des}^* L_p + 1}{L_\xi T_{R,des}^*} = \frac{1}{L_\xi}\left(L_P + \frac{1}{T_{R,des}^*}\right)$$

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According to exercise:

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$$T^*_{R,des} = 1{,}4$$

$$k_{\xi p} = \frac{1}{L_\xi}\left(L_P + \frac{1}{T^*_{R,des}}\right) = \frac{1}{-14{,}3}\left(-0{,}65 + \frac{1}{1{,}4}\right) = -4{,}495 \cdot 10^{-3}$$

Exercise: Basic Controller for the Lateral Motion

The following state space model of a generic aircraft (Class II) in horizontal cruise flight with $V_0 = 150\frac{m}{s}, h = 1000m$) is given:

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a) Determine the control structure and the corresponding controller gains such that the Roll Mode of the aircraft fulfills level 1 handling qualities according to MIL-F-8785C.

b) Determine the handling qualities level according to MIL-F-8785C for the Dutch Roll.

c) Propose a controller structure that changes the relative damping of the Dutch Roll.

d) Determine the gain which can be used to achieve a general, desired relative damping. You can assume that the eigenfrequency of the Dutch Roll remains unchanged by the feedback control.

Determine the handling qualities level according to MIL-F-8785C for the Dutch roll.

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} -0.87 & 6.47 & -0.41 & 0 \\ -1 & -0.38 & 0 & 0.07 \\ 0.91 & -18.8 & -0.65 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} -0.88 & -7.78 \\ 0 & 0.12 \\ -14.3 & 3.74 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$



Note down the Dutch roll as 2x2 state space model (pure yaw oscillation):

Determine the handling qualities level according to MIL-F-8785C for the Dutch roll.

$$\begin{bmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{bmatrix} = \begin{bmatrix} -0.87 & 6.47 & -0.41 & 0 \\ -1 & -0.38 & 0 & 0.07 \\ 0.91 & -18.8 & -0.65 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} r \\ \beta \\ p \\ \Phi \end{bmatrix} + \begin{bmatrix} -0.88 & -7.78 \\ 0 & 0.12 \\ -14.3 & 3.74 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \zeta \end{bmatrix}$$



Note down the Dutch roll as 2x2 state space model (pure yaw oscillation):

$$\begin{pmatrix} \dot{r} \\ \dot{\beta} \end{pmatrix} = \begin{bmatrix} N_r & N_\beta \\ -1 & Y_\beta \end{bmatrix} \cdot \begin{pmatrix} r \\ \beta \end{pmatrix} + \begin{bmatrix} N_\zeta \\ Y_\zeta \end{bmatrix} \cdot \zeta$$

$$N_r = -0.87$$

$$N_\beta = 6.47$$

$$Y_\beta = -0.38$$

Handling qualities Dutch roll:

Level	Phase	Class	Min ζ_d*	Min $\zeta_d\omega_{0d}$ * rad/sec	Min ω_{0d} rad/sec
1	A	II, III	0,19	0,35	0,4
	B	All	0,08	0,15	0,4
	C	II-L, III	0,08	0,10	0,4
2	All	All	0,02	0,05	0,4
3	All	All	0	-	0,4

Requirements for relative damping ζ_d , absolute damping $\sigma = -\zeta_d\omega_{0d}$ and eigenfrequency ω_{0d}

$$\begin{pmatrix}\dot{r}\\ \dot{\beta}\end{pmatrix} = \begin{bmatrix}N_r & N_\beta\\ -1 & Y_\beta\end{bmatrix} \cdot \begin{pmatrix}r\\ \beta\end{pmatrix} + \begin{bmatrix}N_\zeta\\ Y_\zeta\end{bmatrix} \cdot \zeta$$

Characteristic equation:

$$\det(sI - A) = \det\begin{bmatrix}s - N_r & -N_\beta\\ 1 & s - Y_\beta\end{bmatrix} = 0$$

$$(s - N_r) \cdot (s - Y_\beta) + N_\beta = 0$$

$$s^2 - (N_r + Y_\beta)s + N_r \cdot Y_\beta + N_\beta = 0$$

Compare coefficients with: $s^2 + 2 \cdot \zeta_d \omega_0 \cdot s + \omega_0^2 = 0$

$$(N_r + Y_\beta) = -2 \cdot \zeta_{d,DR} \cdot \omega_{0,DR}$$

$$N_r \cdot Y_\beta + N_\beta = \omega_{0,DR}^2$$

Compare coefficients:

$$(N_r + Y_\beta) = -2 \cdot \zeta_{d,DR} \cdot \omega_{0,DR}$$

$$N_r \cdot Y_\beta + N_\beta = \omega_{0,DR}^2$$

$$\omega_{0,DR}^2 = -0.87 \cdot (-0.38) + 6.47 \rightarrow \omega_{0,DR} = 2{,}61 \frac{rad}{s}$$

$$\zeta_{d,DR} = -\frac{1}{2} \cdot \frac{(N_r + Y_\beta)}{\omega_{0,DR}} = -\frac{1}{2} \cdot \frac{(-0{,}87) + (-0{,}38)\ )}{2{,}61} = 0{,}24$$

Handling qualities Dutch roll:

Requirements for relative damping ζ_d and eigenfrequency ω_{0d}

Level	Phase	Class	Min ζ_d*	Min $\zeta_d\omega_{0d}$ * rad/sec	Min ω_{0d} rad/sec
1	A	II, III	0,19	0,35	0,4
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2	All	All	0,02	0,05	0,4
3	All	All	0	-	0,4

$$\omega_{0,DR} = 2{,}61\frac{rad}{s} > 0{,}4$$

$$\zeta_{d,DR} = 0{,}24 > 0.08$$

$$\omega_{0,DR} \cdot \zeta_{d,DR} = 0.6264 > 0{,}15$$

According to exercise:

- generic aircraft (Class II)
- in horizontal cruise flight
- level 1 desired

Level 1 is satisfied!

Exercise: Basic Controller for the Lateral Motion

The following state space model of a generic aircraft (Class II) in horizontal cruise flight with $V_0 = 150\frac{m}{s}, h = 1000m$) is given:

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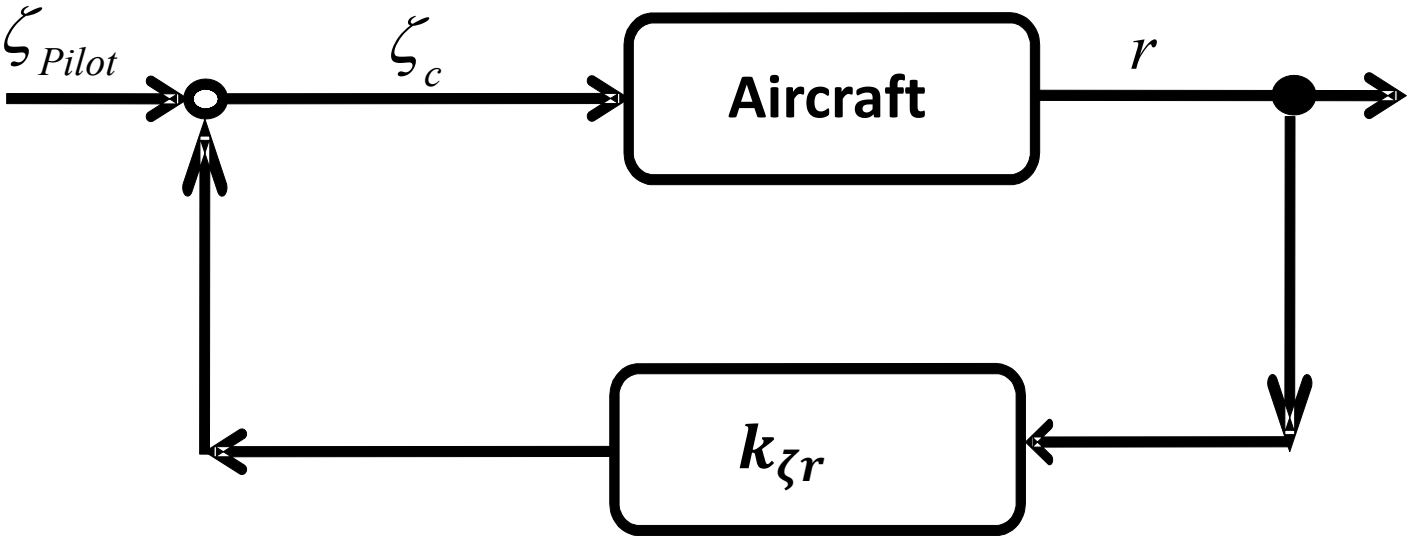
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c) Propose a controller structure that changes the relative damping of the Dutch Roll.

d) Determine the gain which can be used to achieve a general, desired relative damping. You can assume that the eigenfrequency of the Dutch Roll remains unchanged by the feedback control.

Propose a controller structure that changes the relative damping of the Dutch roll.



Yaw rate feedback

Exercise: Basic Controller for the Lateral Motion

The following state space model of a generic aircraft (Class II) in horizontal cruise flight with $V_0 = 150\frac{m}{s}, h = 1000m$) is given:

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d) Determine the gain which can be used to achieve a general, desired relative damping. You can assume that the eigenfrequency of the Dutch Roll remains unchanged by the feedback control.

Determine the gain which leads to a **desired relative damping**. Assume that the eigenfrequency of the Dutch Roll remains unchanged for the feedback control with the yaw rate.

$$\begin{pmatrix}\dot{r}\\ \dot{\beta}\end{pmatrix} = \begin{bmatrix}N_r & N_\beta\\ -1 & Y_\beta\end{bmatrix} \cdot \begin{pmatrix}r\\ \beta\end{pmatrix} + \begin{bmatrix}N_\zeta\\ 0\end{bmatrix} \cdot \zeta$$

Control law: $\zeta_c = \zeta_{Pilot} - k_{\zeta r} \cdot r$



Derive the closed loop dynamics:

Insert control law in state space model:

$$\begin{pmatrix}\dot{r}\\ \dot{\beta}\end{pmatrix} = \begin{bmatrix}N_r & N_\beta\\ -1 & Y_\beta\end{bmatrix} \cdot \begin{pmatrix}r\\ \beta\end{pmatrix} + \begin{bmatrix}N_\zeta\\ 0\end{bmatrix} \cdot (\zeta_{Pilot} - k_{\zeta r} \cdot r)$$

$$\begin{pmatrix}\dot{r}\\ \dot{\beta}\end{pmatrix} = \begin{bmatrix}N_r - k_{\zeta r} \cdot N_\zeta & N_\beta\\ -1 & Y_\beta\end{bmatrix} \cdot \begin{pmatrix}r\\ \beta\end{pmatrix} + \begin{bmatrix}N_\zeta\\ 0\end{bmatrix} \cdot \zeta_{Pilot}$$

Artificial N_r^*: $N_r^* = N_r - k_{\zeta r} \cdot N_\zeta$

from 12 b:

$$\left(N_r^* + Y_\beta\right) = -2 \cdot \zeta_{d,DR}^* \cdot \omega_{0,DR}^*$$

$$N_r^* \cdot Y_\beta + N_\beta = \omega_{0,DR}^{*2}$$

Assumption from 12 d: $\omega_{0,DR}^* \approx \omega_{0,DR}$

$$N_r = -0.87$$
$$N_\beta = 6.47$$
$$Y_\beta = -0.38$$

$$\omega_{0,DR}^* = \omega_{0,DR} = \sqrt{N_r \cdot Y_\beta + N_\beta}$$

N_r without *!!

$$\zeta_{d,DR}^* = -\frac{1}{2}\frac{\left(N_r^* + Y_\beta\right)}{\omega_{0,DR}}$$

Insert $N_r^* = N_r - k_{\zeta r} \cdot N_\zeta$ in $\zeta_{d,DR}^* = -\frac{1}{2}\frac{\left(N_r^* + Y_\beta\right)}{\omega_{0,DR}}$

$$\zeta_{d,DR}^* = -\frac{1}{2}\frac{\left(N_r - k_{\zeta r} \cdot N_\zeta + Y_\beta\right)}{\omega_{0,DR}}$$



Solve for $k_{\zeta r}$

$$\zeta_{d,DR}^* = -\frac{1}{2} \cdot \frac{N_r + Y_\beta}{\omega_{0,DR}} + \frac{1}{2} \cdot \frac{k_{\zeta r} \cdot N_\zeta}{\omega_{0,DR}}$$

$$\zeta_{DR}^* = \zeta_{DR} + \frac{1}{2} \cdot \frac{k_{\zeta r} \cdot N_\zeta}{\omega_{0,DR}}$$

$$k_{\zeta r} = \frac{2 \cdot \omega_{0,DR} \cdot (\zeta_{DR}^* - \zeta_{DR})}{N_\zeta}$$